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The percentage error in x is given by

$$\frac{\Delta x}{x}\times 100 = 2\frac{\Delta a}{a}\times 100 + 3\frac{\Delta b}{b}\times 100$$

$$+\frac{\Delta c}{c}\times 100 + \frac{1}{2}\frac{\Delta d}{d}\times 100$$

$$= (2\times 4\%) + (3\times 2\%) + (1\times 3\%) + (\tfrac{1}{2}\times 1\%)$$

$$= 8\% + 6\% + 3\% + 0.5\%$$

The percentage error is $x = 17.5\%$

1.7 SIGNIFICANT FIGURES

1.7.1 Definition and Rules of Significant Figures

The digits which tell us the number of units we are reasonably sure of having counted in making a measurement are called significant figures.

The digits that are known reliably plus the first uncertain digit are known as significant figures or significant digits.

For example, the value of gravitational constant is 6.67×10^{-11} N m^2 kg^{-2}. Here the digits 6 and 6 are reliable and certain, while the digit 7 is uncertain. Thus the measured value has three significant figures.

The rules for counting significant figures are given in Table 1.9.

EXAMPLE 1.10

State the number of significant figures in the following

i) 600800
ii) 400
iii) 0.007
iv) 5213.0
v) $2.65\times 10^{24}\,m$
vi) 0.0006032

Solution: i) four ii) one iii) one iv) five v) three vi) four

1.7.2 Rounding Off

Calculators are widely used now-a-days to do calculations. The result given by a calculator has too many figures. In no case should the result have more significant figures than the figures involved in the data used for calculation. The result of calculation with numbers containing more than one uncertain digit should be rounded off. The rules for rounding off are shown in Table 1.10.

EXAMPLE 1.11

Round off the following numbers as indicated

i) 18.35 up to 3 digits
ii) 19.45 up to 3 digits
iii) 101.55×10^6 up to 4 digits
iv) 248337 up to digits 3 digits
v) 12.653 up to 3 digits.

Solution

i) 18.4 ii) 19.4 iii) 101.6×10^6
iv) 248000 v) 12.7